Clouded Judgement 12.12.25 - Long Live Systems of Record
云雾判断 12.12.25 - 报表系统永存

JAMIN BALL
DEC 12, 2025 2025 年 12 月 12 日

149 9 11 Share 分享

Every week I'll provide updates on the latest trends in cloud software companies Follow along to stay up to date!

每周我都会提供关于云软件公司最新趋势的更新。关注我以保持最新动态！

readlater.rwpi4p18mji@instapa Subscribe 订阅

Systems of Record are Dead. Long Live Systems of Record

记录系统已死。记录系统万岁

Every few weeks I see some version of the same claim: "The new system of recor the agent," or "Workflows swallow systems of record," or "Data is the system of r apps are just thin views." There is a grain of truth in each of those, but I think it easy to over-rotate and accidentally throw out the thing enterprises still need mo which is not a "system of record" as much as a reliable source of truth.

每隔几周，我都会看到类似的说法："新的记录系统是代理"，或"工作流吞噬了记 统"，或"数据是记录系统，应用只是简单的视图"。这些说法中都含有一些真相， 认为也很容易过度解读，从而无意中抛弃了企业仍然最需要的东西——这与其说是 "记录系统"，不如说是一个可靠的事实来源。

When I think about systems of record, I do not actually think in product categori think in the much more boring lens of "where does the truth live." In other word enterprise workflow needs to know something at a specific step, where is the one

that answer is considered canonical. Because as workflows get more automated a more agent driven, the fragility point often has nothing to do with the model anc everything to do with whether the agent pulled the right value from the right sys the right time. If an early step in a quote to cash flow grabs the wrong price list, wrong contract term, or a stale ARR number, the rest of the workflow is now confidently automating the wrong thing.

当我思考记录系统时，我实际上并不从产品类别去思考。我更关注的是一个相对 视角："真相存在于何处"。换句话说，如果企业工作流程在特定步骤需要了解某 那么哪个地方的答案被认为是权威的？因为随着工作流程的自动化和代理驱动程 越高，脆弱点通常与模型本身无关，而完全取决于代理是否在正确的时间从正确 中提取了正确的数值。如果在报价到收款流程的早期步骤中获取了错误的价格列 误的合同条款或过时的年经常性收入（ARR）数字，那么后续的工作流程就会自 动化错误的内容。

Anyone who has spent time inside a large company knows how messy this gets i practice. Take something as simple as "what is our ARR." Ask the sales org and y will get one number. Ask finance and you get another, with a different set of exclusions and adjustments. Ask accounting, now you are talking revenue recogr not bookings. Ask legal, and they will correctly remind you that half the "ARR" i fast growing business is backed by contracts that look nothing like the neat recu subscription you want it to be. Even the definition is slippery. For a consumptior business, is ARR the annualized run rate from last month's usage, is it contracte commit, is it contracted commit net of discounts and credits, or is it a lagging tw month billing number dressed up to sound like a recurring metric.

任何在大型企业工作过的人，都知道实际情况有多么混乱。就拿一个简单的问题 "我们的年度经常性收入（ARR）是多少？"向销售部门询问，会得到一个数字；向 部门询问，则会得到另一个数字，而且排除项和调整项也各不相同。向会计部门 现在讨论的就变成了收入确认，而不是预订。向法务部门询问，他们会正确地提 在高速增长的业务中，一半的"ARR"是基于那些与你所期望的整齐的定期订阅截 的合同。甚至连定义都是模糊的。对于按需付费的业务而言，ARR 是指上个月使 的年化运行速率吗？是合同承诺金额吗？是扣除折扣和信用后的合同承诺金额，这 过去十二个月的开票额为基础，包装成看起来像定期指标的滞后数据？

Now imagine you tell an agent, “Go calculate ARR by segment and send a deck t board.” Which ARR should it use. Which table is canonical. If sales and finance disagree, who wins. If the billing system and the warehouse have drifted by a few percent, which one does the agent treat as truth. This is the part of the “systems record are dead” argument that feels off to me. The more we automate, the more important it becomes that someone has done the unglamorous work of deciding the correct answer is and where it lives.

现在想象一下，你告诉一个代理：“去按部门计算年度经常性收入（ARR），并发份幻灯片给董事会。”它应该使用哪个 ARR？哪个表格是权威的？如果销售和财务意见不一致，谁说了算？如果计费系统和仓库数据偏差了几个百分点，代理应该以作为真相？这正是“信息系统纪要已死”这一论点中让我觉得不妥之处。自动化程度高，就越需要有人完成那些乏味的工作，即确定正确答案是什么，以及它存在于何

Historically, systems of record solved this in a fairly straightforward way. You ha CRM system of record for customers and opportunities, an ERP for financials, ar HRIS for people, a billing system for invoices, and so on. They were not perfect, each domain had a primary home. Then the last decade arrived and everyone trie centralize that reality into the warehouse or lakehouse. The pitch was that if you poured all the data into one place and layered semantic models and metrics defir on top, you would finally get a single source of truth that analytics, dashboards a downstream tools could all agree on.

历史上，信息系统纪要以一种相当直接的方式解决了这个问题。你有一个用于客户机的 CRM 系统作为记录，一个用于财务的 ERP 系统，一个用于人员的 HRIS 系统个用于发票的计费系统，等等。它们并不完美，但每个领域都有一个主要的归属地后，在过去十年中，每个人都试图将这种现实集中到数据仓库或湖仓中。其论点是果你将所有数据汇集到一个地方，并在其上叠加语义模型和指标定义，你最终将获个单一的真相来源，分析、仪表板和下游工具都可以达成一致。

In practice, that vision got part of the way there. Warehouses and lakehouses absolutely became the gravitational center for analytical truth in many companie Teams built careful dbt models, they put governance around “official” metrics, th curated gold tables that fed BI tools. The problem is that most of this lived

downstream of the operational world. The sales team still lived in Salesforce, the finance team still closed the books in NetSuite, the support team still worked tic in Zendesk. The warehouse or lakehouse was the retrospective mirror, not the transactional front door.

实际上，这一愿景在一定程度上得到了实现。在许多公司中，数据仓库和数据湖仓成为了分析真相的引力中心。团队构建了谨慎的 dbt 模型，围绕“官方”指标建立了机制，并维护了向 BI 工具提供数据的黄金表。但问题在于，绝大多数这些工作都运营世界的下游。销售团队仍然生活在 Salesforce 中，财务团队仍在 NetSuite 中账目结算，支持团队仍在 Zendesk 中处理工单。数据仓库或数据湖仓只是回顾性像，而非事务性操作的前端入口。

Agents change that equation in two important ways. First, they are inherently cr system. When you tell an agent to “run a quote to cash workflow,” you are asking dance across CRM, CPQ, billing, collections, and maybe a few internal tools that in random corners of the org. Second, they are inherently action oriented. This is just about running a report or producing a dashboard, it is about taking actions t change state in those underlying systems. That combination means agents are or good as their understanding of which system owns which truth, and what the co is between those truths. This is the bull thesis on a company like Databricks - th become the center of gravity for AI Agents, and start to build these agents thems

代理在两个重要方面改变了这一方程式。首先，它们本质上是跨系统的。当你告诉代理“运行从询价到收款的工作流程”时，你实际上是要求它在 CRM、CPQ、计费收、以及可能存在于组织内各个角落的几个内部工具之间协同工作。其次，它们本是行动导向的。这不仅仅是运行报告或生成仪表板，而是采取能够改变底层系统状操作。这种组合意味着代理的能力仅取决于它们对哪个系统拥有何种真相的了解，这些真相之间的契约关系。这就是像 Databricks 这样的公司的核心论点——它们为 AI 代理的重心，并开始自行构建这些代理。

Said another way, agents are forcing us to separate the UX of work from the sou truth for work. The UX can now be a chat window, a natural language interface, specialized agent UI that looks nothing like the old enterprise applications. Und hood, though, something still has to say “this is the canonical customer record” o

"this is the legally binding contract term" or "this number is the one we report to Street." That "something" might be a traditional system of record, it might be a warehouse-backed semantic layer, or it might be a new class of "data control plan product, but it is absolutely not going away.

换句话说，代理正在迫使我们把工作的用户体验与工作的真实来源分离开来。用户现在可以是一个聊天窗口、自然语言界面，或是一个与旧企业应用完全不同的专用用户界面。但在后台，仍然需要某种东西来声明"这是权威的客户记录"或"这是具律约束力的合同条款"或"这个数字是我们向华尔街报告的"。这个"某种东西"可能的记录系统，也可能是基于数据仓库的语义层，或者可能是一种全新的"数据控制产品，但它绝不会消失。

This is where I think data warehouses and lakehouses start to look like the natur substrate for many agentic workflows. They already try to centralize data across domains. They already carry the semantic layer that defines metrics and business entities. They already sit at the intersection of governance, lineage, and access co If you squint a bit, the warehouse plus the metrics layer plus the governance tool starts to look less like a reporting system and more like a "truth registry" that the of the company can rely on.

这正是我认为数据仓库和数据湖仓开始看起来像许多代理工作流的天然基础平台的因。它们已经尝试在不同领域中集中数据，已经承载了定义指标和业务实体的语义也已经位于治理、血缘和访问控制的交叉点上。如果从远处观察，数据仓库加上指和治理工具，开始看起来不再像一个报告系统，而更像是一个公司其他部分可以信"真相注册表"。

The missing piece is that most of these stacks were designed for humans running queries, not agents orchestrating workflows. Humans can hold nuance in their h If finance and sales disagree on ARR, they can talk it out and decide which numb use for the board deck. Agents cannot. They need explicit rules. They need confli resolution baked into the data model. They need to know that "official_arr" is th metric we use for external reporting, "sales_arr" is the metric we use for comp pl and "product_arr" is the metric we use for feature level analysis. They need the warehouse to not only store the data but also encode the precedence and meanin

that data.

缺失的一环是，这些系统大部分是为人类执行查询而设计的，而不是为代理协调工
而设计的。人类可以在脑海中把握细微差别。如果财务和销售对年经常性收入（A
存在分歧，他们可以进行讨论并决定在董事会演示文稿中使用哪个数字。而代理无
到这一点。它们需要明确的规则。它们需要在数据模型中内置冲突解决机制。它们
知道“official_arr”是我们用于外部报告的指标，“sales_arr”是我们用于薪酬计划的
标，而“product_arr”是我们用于功能级别分析的指标。它们需要数据仓库不仅存储
据，还要编码数据的优先级和含义。

There is a similar story on the operational side. I do not think CRM or ERP or bi
systems disappear. Instead, they quietly evolve into something closer to “state
machines with APIs” that are optimized for programmatic access rather than hu
UI. The agent initiates the quote, computes the pricing, assembles the contract,
negotiates redlines, and then, at the right moment, writes the final state into the
downstream system of record through a well defined boundary. The human migh
see that state in a web interface, but the primary consumer is an agent. The syste
record becomes less of a place you click around in and more of a durable storage
constraint engine that machines talk to.

在运营侧也有类似的故事。我不认为 CRM、ERP 或计费系统会消失，相反，它们
悄然演变为更接近“带有 API 的状态机”的形式，其优化方向是程序化访问，而非人
类的用户界面。Agent 会发起报价、计算价格、组装合同、协商条款，然后在恰当
机，通过定义明确的边界将最终状态写入下游的系统记录中。人类可能依然能在网
面上看到该状态，但主要的消费者是 Agent。系统记录不再是一个你点击操作的地
而更像是机器之间通信的持久存储和约束引擎。

If you believe that world, then the question “are systems of record dying” starts t
a bit like asking “is the database dying” when people moved from client server ap
web apps. The form factor is changing. The power center may migrate from
monolithic SaaS fronts to shared data layers and workflows. But the need for a
contract that says “this is the truth, and here is how you are allowed to change it

increases.

如果你相信这个世界，那么“记录系统是否正在消亡”这个问题，开始听起来有点像从客户端/服务器应用转向网页应用时，问“数据库是否正在消亡”。形式在改变。核心可能会从单一的 SaaS 前端迁移到共享的数据层和工作流程。但对一份声明“这是相，以及你如何被允许更改它”的契约需求只会增加。

You can already see this shift in how people talk about “AI native” apps. The mo interesting ones rarely start by building yet another end user UI. They start by si next to the warehouse, or next to the CRM, or next to the ERP, and then building opinionated agents that orchestrate work across those systems. Their edge is usu combination of workflow design, semantic modeling, and tight integration with customer’s existing sources of truth. Underneath the marketing, they are basicall wrapping the messy reality of enterprise data in a cleaner contract so agents can operate safely.

你已经可以观察到人们对“原生 AI”应用的谈论方式正在发生变化。最有趣的项目从打造另一个用户界面开始，而是从与仓库、CRM 或 ERP 系统相邻的位置入手，构建能够协调这些系统之间工作的具有明确立场的代理。它们的优势通常在于工作设计、语义建模以及与客户现有数据源的紧密集成。在营销宣传之下，本质上它们将企业数据混乱的现实封装成一个更清晰的契约，从而使代理能够安全地运行。

There is also a valuation angle hiding in here. Investors love to debate whether “systems of record” get higher multiples than “systems of engagement,” and whe pure workflow tools land in that spectrum. Agents complicate that framing. An a that sits on top of someone else’s data and can be easily unplugged probably does deserve system of record type durability. An agent platform that becomes the pla where metric definitions live, where entity schemas are defined, and where polic about who can do what are enforced, starts to look much more like a source of tr even if the underlying bits are stored in Snowflake or Databricks or some operati database. The multiple will follow the stickiness of the truth, not the buzzword c

slide.

这里还隐藏着估值的角度。投资者们热衷于争论“记录系统”是否比“交互系统”获得的倍数，以及纯工作流工具在这一谱系中的位置。但代理（agent）让这种框架变杂。一个位于他人数据之上的代理，且可以轻松断开连接，可能并不值得享有“记统”级别的耐用性。然而，一个代理平台，若成为度量标准定义、实体模式定义以限策略（谁可以做什么）执行的中心，即使底层数据存储在 Snowflake 或 Databr或某些操作型数据库中，也开始看起来更像一个真相的来源。倍数将追随真相的而不是幻灯片上的流行语。

So where do I land. I do not think systems of record are dying. I think they are ge unbundled and rewired. The “record” part, the actual truth, will increasingly live combination of warehouses, lakehouses, and still important operational systems. top of that, we will get a new layer of semantic contracts and control planes that agents how to safely read and write that truth. The familiar SaaS front ends that to sit on top of those systems of record will matter less over time. Agents and workflow UIs will become the primary way humans interact with work. But the underlying need for a well defined source of truth, with clear ownership and constraints, will only grow.

那么我站在哪里呢？我不认为记录系统正在消亡。我认为它们正在被拆解和重新其中的“记录”部分，即真实的信息，将越来越多地存在于数据仓库、数据湖仓以及重要的运营系统中。在此基础上，我们将出现新的语义契约和控制平面，指导代理安全地读取和写入这些真相。那些曾经位于这些记录系统之上的熟悉的 SaaS 前端着时间的推移将变得不那么重要。代理和工作流 UI 将成为人类与工作交互的主要式。但对一个定义清晰、责任明确且有约束条件的真相来源的底层需求只会不断

Said another way, agents are not replacing systems of record. They are raising th standards for what a good one looks like. The companies that win this cycle will ones that build amazing agentic experiences on top of boring, rock solid sources truth, rather than pretending those sources no longer matter.

换句话说，智能体并没有取代核心系统。它们只是提高了优秀核心系统应有的标这个周期中胜出的公司将是在枯燥但坚如磐石的真相源之上构建出色智能体体验的司，而不是假装这些源不再重要。